Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Wendell Trinkler Jr. Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
1001 W Monte Vista RD	Ceres	Stanislaus	95307

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X008-X016-XXXX

B. OPERATORS

Trinkler, Wendel Jr.

Operator name: Trinkler, Wendel Jr.

Telephone no.: (209) 537-9883 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
7136 Crows Landing RD	Ceres	CA	95307

This operator is responsible for paying permit fees.

C. OWNERS

Trinkler, Wendel Jr.

Legal owner name: Trinkler, Wendel Jr.

Telephone no.: (209) 537-9883 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
7136 Crows Landing RD	Ceres	CA	95307

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	50	0	0	0	0
Number under roof	350	0	0	0	0	0
Maximum number	450	100	0	0	0	0
Average number	350	50	0	0	0	0
Avg live weight (lbs)	1,000	1,100	0	0		

Predominant milk cow breed: Jersey

Average milk production: 55 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 9,018.38 *tons per reporting period*

Total nitrogen from manure: 118,452.12 *lbs per reporting period*

After ammonia losses (30% loss applied): 82,916.48 *lbs per reporting period*

Total phosphorus from manure: 19,595.48 *lbs per reporting period*

Total potassium from manure: 63,578.98 *lbs per reporting period*

Total salt from manure: 176,295.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 23,961,289 *gallons*

Total nitrogen generated: 53,437.82 *lbs*

Total phosphorus generated: 10,427.56 *lbs*

Total potassium generated: 78,083.81 *lbs*

Total salt generated: 731,655.01 *lbs*

	23,961,289	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	23,961,289	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Foy #1	37	37	2	both	X022-X008-X017-XXXX
Foy #2	30	30	0	none	X022-X008-X017-XXXX
Lentz	19	19	0	none	X022-X023-X018-XXXX X022-X023-X019-XXXX
Monte Vista	63	63	2	both	X022-X008-X016-XXXX
Presson	39	39	2	both	X022-X008-X022-XXXX
Sammons	19	19	0	none	X022-X008-X020-XXXX
Stans House	12	12	2	both	X022-X008-X013-XXXX
Totals for areas that were used for application	151	151	8		
Totals for areas that were not used for application	68	68	0		
Land application area totals	219	219	8		

B. CROPS AND HARVESTS

Foy #1

Field name: Foy #1

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	706.75 *ton*	As-is		77.8	4,900.00	900.00	7,500.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.10	187.19	34.38	286.52	1,068.61

Foy #1

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,095.20 *ton*	As-is		68.6	3,900.00	700.00	5,300.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.60	230.88	41.44	313.76	1,040.97

Monte Vista

Field name: Monte Vista

10/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 63 Plant date: 10/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	1,215.90 *ton*	As-is		79.0	4,400.00	900.00	8,300.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.30	169.84	34.74	320.38	997.04

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/02/2023	1,959.30 *ton*	As-is		67.0	4,000.00	700.00	5,300.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.10	248.80	43.54	329.66	1,211.03

Presson

Field name: Presson

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	729.30 *ton*	As-is		72.2	5,700.00	1,200.00	11,000.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.70	213.18	44.88	411.40	1,278.86

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,194.83 *ton*	As-is		68.6	3,900.00	700.00	5,300.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.64	238.97	42.89	324.75	1,077.43

Stans House

Field name: Stans House

10/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 10/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	232.39 *ton*	As-is		76.6	5,000.00	1,000.00	8,600.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.37	193.66	38.73	333.09	1,087.59

Stans House

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	376.81 *ton*	As-is		66.8	4,300.00	700.00	5,100.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.40	270.05	43.96	320.29	1,209.31

NUTRIENT BUDGET

A. LAND APPLICATIONS

Foy #1 - 10/03/2022: Oats, silage-soft dough

Field name: Foy #1

Crop: Oats, silage-soft dough — Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.53	9.66	46.39	606.31	625,175.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.42	6,040,583.00 *gal*
Application event totals		46.53	9.66	46.39	641.73	

12/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.20	11.67	56.03	732.29	755,080.00 *gal*
Application event totals		56.20	11.67	56.03	732.29	

02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.84	9.39	61.87	336.46	500,600.00 *gal*
Application event totals		38.84	9.39	61.87	336.46	

03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.83	12.78	84.15	457.62	680,875.00 *gal*
Canal	Surface water	0.00	0.00	0.00	27.58	4,703,255.00 *gal*
Application event totals		52.83	12.78	84.15	485.20	

Foy #1 - 05/17/2023: Corn, silage

Field name: Foy #1

Crop: Corn, silage — Plant date: 05/17/2023

Foy #1 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.45	6.24	64.74	754.00	875,150.00 *gal*
Canal	Surface water	0.00	0.00	0.00	34.67	5,912,045.00 *gal*
Application event totals		41.45	6.24	64.74	788.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	125.44	47.94	30.02	1,702.40	370.00 *ton*
Application event totals		125.44	47.94	30.02	1,702.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	30.60	5,217,814.00 *gal*
Application event totals		0.00	0.00	0.00	30.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.13	5.14	53.30	620.80	720,550.00 *gal*
Canal	Surface water	0.00	0.00	0.00	28.26	4,819,065.00 *gal*
Application event totals		34.13	5.14	53.30	649.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.14	4,628,256.00 *gal*
Application event totals		0.00	0.00	0.00	27.14	

Foy #1 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.27	6.81	70.71	823.55	955,875.00 *gal*
Canal	Surface water	0.00	0.00	0.00	26.21	4,469,045.00 *gal*
Application event totals		45.27	6.81	70.71	849.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.90	5.70	59.20	689.47	800,255.00 *gal*
Canal	Surface water	0.00	0.00	0.00	22.91	3,907,116.00 *gal*
Application event totals		37.90	5.70	59.20	712.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	22.44	3,826,130.00 *gal*
Application event totals		0.00	0.00	0.00	22.44	

Monte Vista - 10/04/2022: Oats, silage-soft dough

Field name: Monte Vista

Crop: Oats, silage-soft dough

Plant date: 10/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.68	10,071,117.00 *gal*
Application event totals		0.00	0.00	0.00	34.68	

Monte Vista - 10/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.94	11.40	54.77	715.89	1,256,870.00 *gal*
Application event totals		54.94	11.40	54.77	715.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.54	10.77	70.96	385.86	977,512.00 *gal*
Application event totals		44.54	10.77	70.96	385.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.40	10.01	65.96	358.68	908,656.00 *gal*
Application event totals		41.40	10.01	65.96	358.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.52	12.22	80.48	437.64	1,108,700.00 *gal*
Canal	Surface water	0.00	0.00	0.00	27.02	7,845,138.00 *gal*
Application event totals		50.52	12.22	80.48	464.66	

Monte Vista - 05/17/2023: Corn, silage

Field name: Monte Vista

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	125.44	47.94	30.02	1,702.40	630.00 *ton*
Application event totals		125.44	47.94	30.02	1,702.40	

Monte Vista - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.12	10,198,881.00 *gal*
Application event totals		0.00	0.00	0.00	35.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	31.22	9,064,881.00 *gal*
Application event totals		0.00	0.00	0.00	31.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.25	9.67	100.35	1,168.65	2,309,601.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.88	8,675,037.00 *gal*
Application event totals		64.25	9.67	100.35	1,198.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.46	4.13	42.89	499.50	987,150.00 *gal*
Canal	Surface water	0.00	0.00	0.00	27.88	8,096,445.00 *gal*
Application event totals		27.46	4.13	42.89	527.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	26.25	7,623,441.00 *gal*
Application event totals		0.00	0.00	0.00	26.25	

Monte Vista - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.59	4.90	50.90	592.79	1,171,520.00 *gal*
Canal	Surface water	0.00	0.00	0.00	20.90	6,067,130.00 *gal*
Application event totals		32.59	4.90	50.90	613.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.97	4.21	43.69	508.84	1,005,620.00 *gal*
Canal	Surface water	0.00	0.00	0.00	21.49	6,240,160.00 *gal*
Application event totals		27.97	4.21	43.69	530.33	

Presson - 10/03/2022: Oats, silage-soft dough

Field name: Presson

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.61	14.45	69.39	906.98	985,755.00 *gal*
Canal	Surface water	0.00	0.00	0.00	36.06	6,482,385.00 *gal*
Application event totals		69.61	14.45	69.39	943.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.54	12.47	82.11	446.51	700,255.00 *gal*
Application event totals		51.54	12.47	82.11	446.51	

Presson - 10/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.54	10.29	67.76	368.48	577,880.00 *gal*
Application event totals		42.54	10.29	67.76	368.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.49	12.21	80.43	437.37	685,915.00 *gal*
Canal	Surface water	0.00	0.00	0.00	27.53	4,948,125.00 *gal*
Application event totals		50.49	12.21	80.43	464.90	

Presson - 05/13/2023: Corn, silage

Field name: Presson

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.19	7.25	75.27	876.64	1,072,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	37.00	6,651,450.00 *gal*
Application event totals		48.19	7.25	75.27	913.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	125.44	47.94	30.02	1,702.40	390.00 *ton*
Application event totals		125.44	47.94	30.02	1,702.40	

Presson - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	33.15	5,958,615.00 *gal*
Application event totals		0.00	0.00	0.00	33.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	30.85	5,545,722.00 *gal*
Application event totals		0.00	0.00	0.00	30.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.02	5.87	60.94	709.76	868,337.00 *gal*
Canal	Surface water	0.00	0.00	0.00	28.07	5,045,235.00 *gal*
Application event totals		39.02	5.87	60.94	737.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.04	4.67	48.48	564.61	690,754.00 *gal*
Canal	Surface water	0.00	0.00	0.00	26.84	4,824,846.00 *gal*
Application event totals		31.04	4.67	48.48	591.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.94	5.26	54.57	635.50	777,485.00 *gal*
Canal	Surface water	0.00	0.00	0.00	21.31	3,829,674.00 *gal*
Application event totals		34.94	5.26	54.57	656.81	

Presson - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	21.70	3,900,614.00 *gal*
Application event totals		0.00	0.00	0.00	21.70	

Stans House - 10/04/2022: Oats, silage-soft dough

Field name: Stans House

Crop: Oats, silage-soft dough

Plant date: 10/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.81	10.76	51.66	675.13	225,775.00 *gal*
Canal	Surface water	0.00	0.00	0.00	36.98	2,045,340.00 *gal*
Application event totals		51.81	10.76	51.66	712.11	

01/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.26	17.48	115.11	625.95	302,050.00 *gal*
Application event totals		72.26	17.48	115.11	625.95	

02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.87	10.85	71.48	388.70	187,565.00 *gal*
Application event totals		44.87	10.85	71.48	388.70	

Stans House - 10/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.28	9.26	60.98	331.62	160,020.00 *gal*
Canal	Surface water	0.00	0.00	0.00	26.69	1,476,036.00 *gal*
Application event totals		38.28	9.26	60.98	358.30	

Stans House - 05/17/2023: Corn, silage

Field name: Stans House

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.61	8.37	86.85	1,011.50	380,765.00 *gal*
Canal	Surface water	0.00	0.00	0.00	36.01	1,991,832.00 *gal*
Application event totals		55.61	8.37	86.85	1,047.51	

05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	125.44	47.94	30.02	1,702.40	120.00 *ton*
Application event totals		125.44	47.94	30.02	1,702.40	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	32.72	1,809,420.00 *gal*
Application event totals		0.00	0.00	0.00	32.72	

Stans House - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.36	6.68	69.29	806.92	303,755.00 *gal*
Canal	Surface water	0.00	0.00	0.00	30.59	1,692,024.00 *gal*
Application event totals		44.36	6.68	69.29	837.52	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	28.86	1,596,360.00 *gal*
Application event totals		0.00	0.00	0.00	28.86	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.82	4.49	46.57	542.36	204,165.00 *gal*
Canal	Surface water	0.00	0.00	0.00	27.64	1,528,944.00 *gal*
Application event totals		29.82	4.49	46.57	570.01	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.08	4.38	45.42	528.97	199,124.00 *gal*
Canal	Surface water	0.00	0.00	0.00	25.47	1,408,473.00 *gal*
Application event totals		29.08	4.38	45.42	554.44	

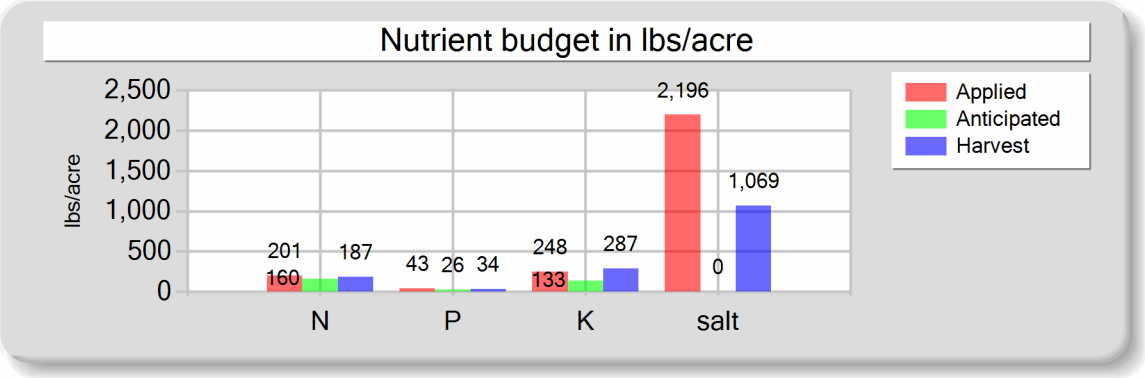
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	25.80	1,426,913.00 *gal*
Application event totals		0.00	0.00	0.00	25.80	

B. NUTRIENT BUDGET

Foy #1 - 10/03/2022: Oats, silage-soft dough

Field name: Foy #1 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.40	43.49	248.44	2,132.69
Fresh water	0.00	0.00	0.00	63.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.40	43.49	248.44	2,195.69
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	187.19	34.38	286.52	1,068.61
Nutrient balance	14.20	9.11	-38.08	1,127.08
Applied to removed ratio	1.08	1.27	0.87	2.05

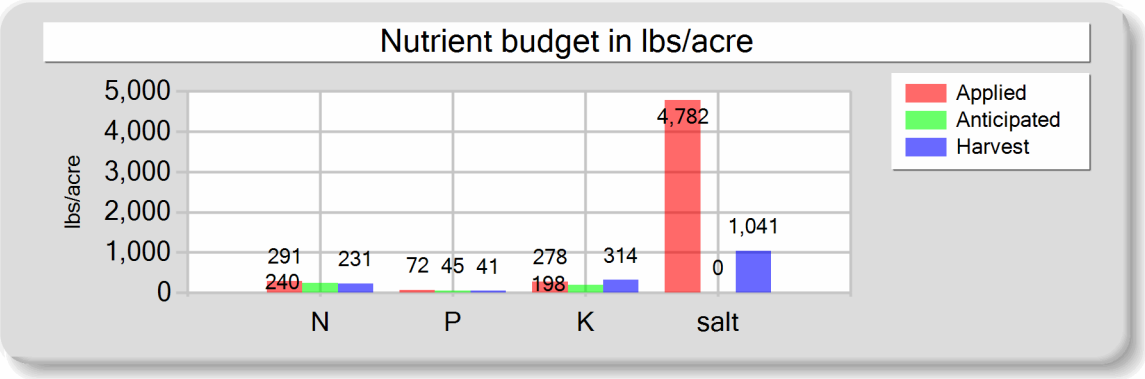
Fresh water applied	
10,743,838.00	*gallons*
395.66	*acre-inches*
10.69	*inches/acre*

Process wastewater applied	
2,561,730.00	*gallons*
94.34	*acre-inches*
2.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Foy #1 - 05/17/2023: Corn, silage

Field name: Foy #1 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	125.44	47.94	30.02	1,702.40
Process wastewater	158.75	23.89	247.96	2,887.82
Fresh water	0.00	0.00	0.00	192.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.19	71.82	277.98	4,782.44
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	230.88	41.44	313.76	1,040.97
Nutrient balance	60.31	30.38	-35.78	3,741.47
Applied to removed ratio	1.26	1.73	0.89	4.59

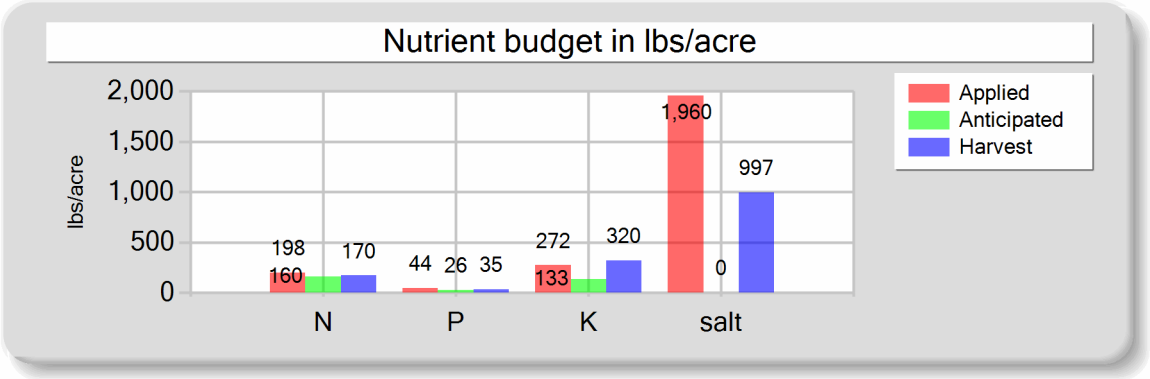
Fresh water applied	
32,779,471.00	*gallons*
1,207.16	*acre-inches*
32.63	*inches/acre*

Process wastewater applied	
3,351,830.00	*gallons*
123.44	*acre-inches*
3.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

Monte Vista - 10/04/2022: Oats, silage-soft dough

Field name: Monte Vista | Crop: Oats, silage-soft dough | Plant date: 10/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.41	44.41	272.17	1,898.06
Fresh water	0.00	0.00	0.00	61.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.41	44.41	272.17	1,959.76
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	169.84	34.74	320.38	997.04
Nutrient balance	28.57	9.67	-48.21	962.72
Applied to removed ratio	1.17	1.28	0.85	1.97

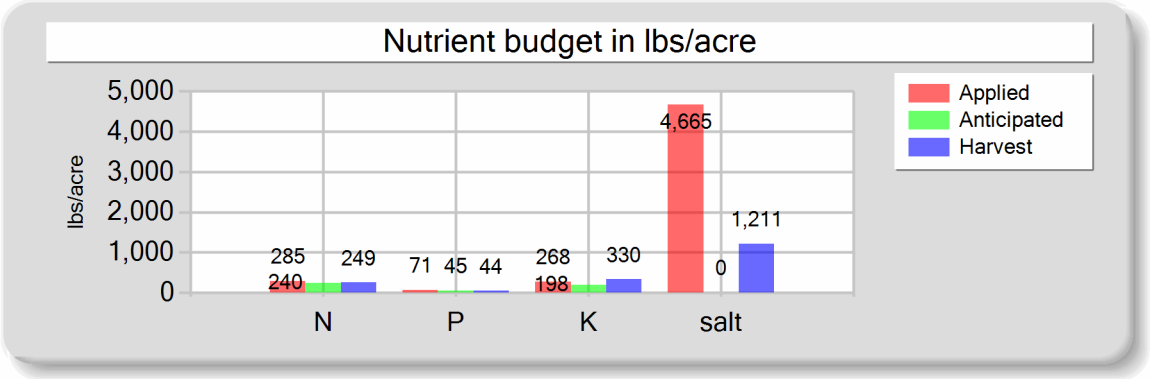
Fresh water applied	
17,916,255.00	*gallons*
659.79	*acre-inches*
10.47	*inches/acre*

Process wastewater applied	
4,251,738.00	*gallons*
156.58	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Monte Vista - 05/17/2023: Corn, silage

Field name: Monte Vista Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	125.44	47.94	30.02	1,702.40
Process wastewater	152.27	22.91	237.82	2,769.78
Fresh water	0.00	0.00	0.00	192.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.71	70.85	267.84	4,664.93
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	248.80	43.54	329.66	1,211.03
Nutrient balance	35.91	27.31	-61.82	3,453.89
Applied to removed ratio	1.14	1.63	0.81	3.85

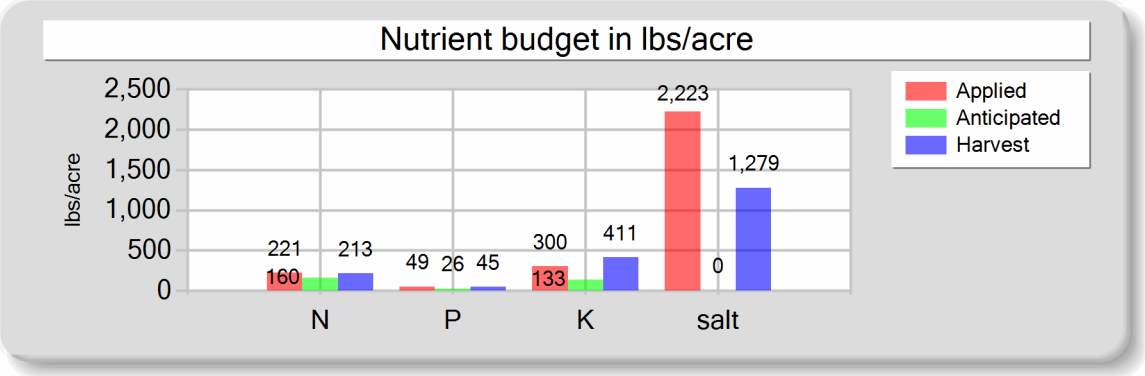
Fresh water applied	
55,965,975.00	*gallons*
2,061.04	*acre-inches*
32.71	*inches/acre*

Process wastewater applied	
5,473,891.00	*gallons*
201.58	*acre-inches*
3.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

Presson - 10/03/2022: Oats, silage-soft dough

Field name: Presson | Crop: Oats, silage-soft dough | Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.17	49.41	299.70	2,159.35
Fresh water	0.00	0.00	0.00	63.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.17	49.41	299.70	2,222.94
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	213.18	44.88	411.40	1,278.86
Nutrient balance	7.99	4.53	-111.70	944.08
Applied to removed ratio	1.04	1.10	0.73	1.74

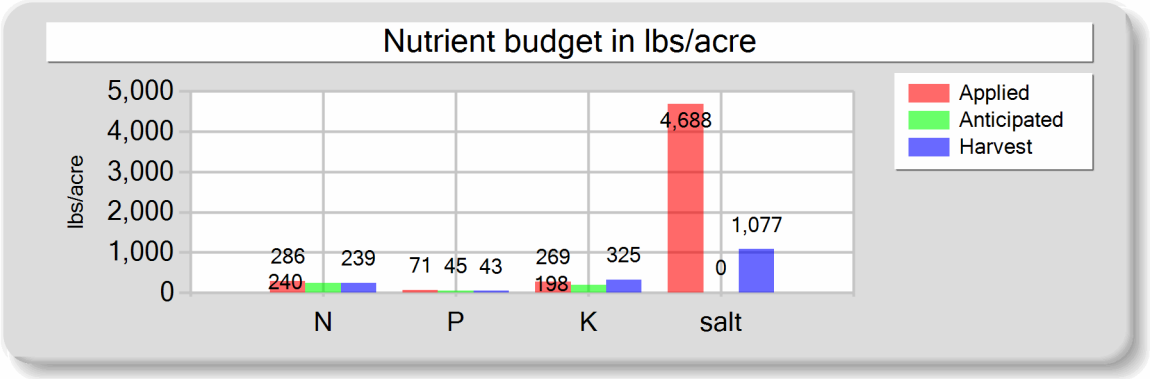
Fresh water applied	
11,430,510.00	*gallons*
420.95	*acre-inches*
10.79	*inches/acre*

Process wastewater applied	
2,949,805.00	*gallons*
108.63	*acre-inches*
2.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

Presson - 05/13/2023: Corn, silage

Field name: Presson Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	125.44	47.94	30.02	1,702.40
Process wastewater	153.19	23.05	239.26	2,786.52
Fresh water	0.00	0.00	0.00	198.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.63	70.99	269.28	4,687.84
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	238.97	42.89	324.75	1,077.43
Nutrient balance	46.66	28.10	-55.47	3,610.41
Applied to removed ratio	1.20	1.66	0.83	4.35

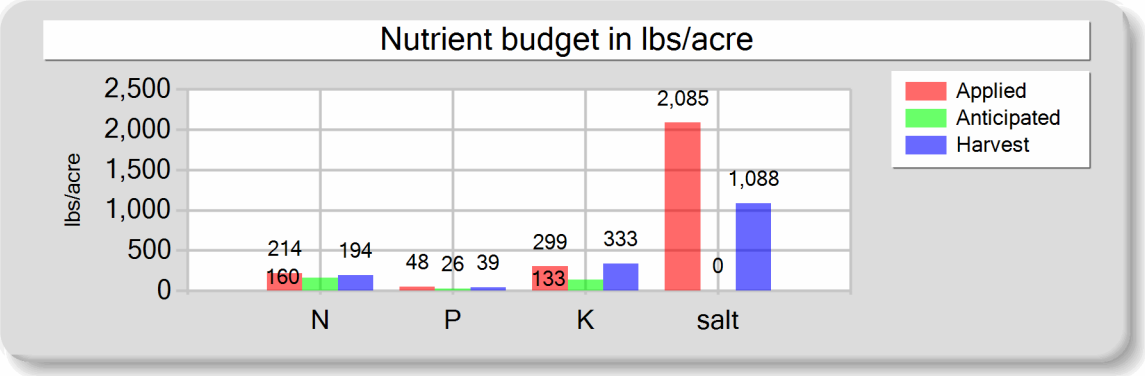
Fresh water applied	
35,756,156.00	*gallons*
1,316.78	*acre-inches*
33.76	*inches/acre*

Process wastewater applied	
3,409,076.00	*gallons*
125.54	*acre-inches*
3.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stans House - 10/04/2022: Oats, silage-soft dough

Field name: Stans House | Crop: Oats, silage-soft dough | Plant date: 10/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.22	48.34	299.22	2,021.40
Fresh water	0.00	0.00	0.00	63.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.22	48.34	299.22	2,085.07
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	193.66	38.73	333.09	1,087.59
Nutrient balance	20.56	9.61	-33.87	997.48
Applied to removed ratio	1.11	1.25	0.90	1.92

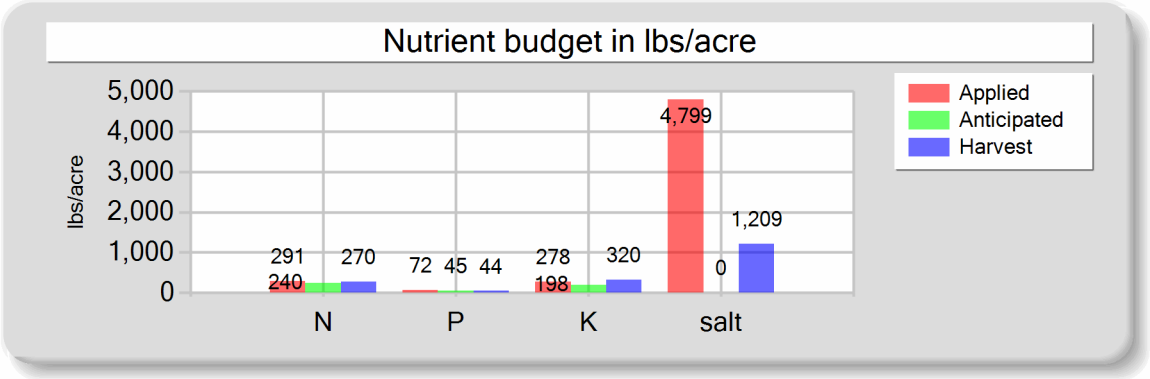
Fresh water applied	
3,521,376.00	*gallons*
129.68	*acre-inches*
10.81	*inches/acre*

Process wastewater applied	
875,410.00	*gallons*
32.24	*acre-inches*
2.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stans House - 05/17/2023: Corn, silage

Field name: Stans House | Crop: Corn, silage | Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	125.44	47.94	30.02	1,702.40
Process wastewater	158.86	23.90	248.13	2,889.76
Fresh water	0.00	0.00	0.00	207.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.30	71.84	278.14	4,799.25
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	270.05	43.96	320.29	1,209.31
Nutrient balance	21.25	27.88	-42.15	3,589.94
Applied to removed ratio	1.08	1.63	0.87	3.97

Fresh water applied	
11,453,966.00	*gallons*
421.81	*acre-inches*
35.15	*inches/acre*

Process wastewater applied	
1,087,809.00	*gallons*
40.06	*acre-inches*
3.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall 2022 Separated Solids

Sample and source description: Fall 2022 Separated Solids

Sample date: 11/15/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	5,000.00	8,600.00							31.20
DL	100.00	100.00	100.00							0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 92.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,200.00	3,500.00	4,700.00							12.30
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,000.00	10,700.00	6,700.00	32,700.00	7,500.00	1,600.00	6,700.00	1,960.00		38.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	330.00	246.00	0.00	0.00	68.50	329.00								5,540.00	4,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	344.00	213.00	0.00	0.00	83.20	548.00								4,580.00	2,980
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	619.00	375.00	0.00	0.00	89.50	1,040.00								7,480.00	6,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	210.00	202.00	0.00	0.00	31.60	328.00	222.00	97.70	137.00	1,920.00	0.00	178.00	243.00	4,380.00	3,820
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	350.00	246.00	0.00	0.00	25.90	571.00								5,190.00	8,100
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

T.I.D Canal

Sample description: T.I.D Canal

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								31.10	26
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

IW #1

Sample description: IW #1

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	60.00	0.00	59.70								1,560.00	860
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

IW #2

Sample description: IW #2

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	50.00	0.00	47.20								1,350.00	750
DL	0.50	0.50	0.50								1.00	1

IW #3

Sample description: IW #3

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	46.00	0.00	46.00								1,370.00	765
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Foy #1 - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	900.00	7,500.00		12.60
DL	100.00	100.00	100.00		0.01

Foy #1 - 05/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	700.00	5,300.00		5.60
DL	100.00	100.00	100.00		0.01

Monte Vista - 10/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	900.00	8,300.00		12.30
DL	100.00	100.00	100.00		0.01

Monte Vista - 05/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	700.00	5,300.00		5.90
DL	100.00	100.00	100.00		0.01

Presson - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,700.00	1,200.00	11,000.00		12.30
DL	100.00	100.00	100.00		0.01

Presson - 05/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	700.00	5,300.00		5.60
DL	100.00	100.00	100.00		0.01

Stans House - 10/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	1,000.00	8,600.00		12.00
DL	100.00	100.00	100.00		0.01

Stans House - 05/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	700.00	5,100.00		5.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

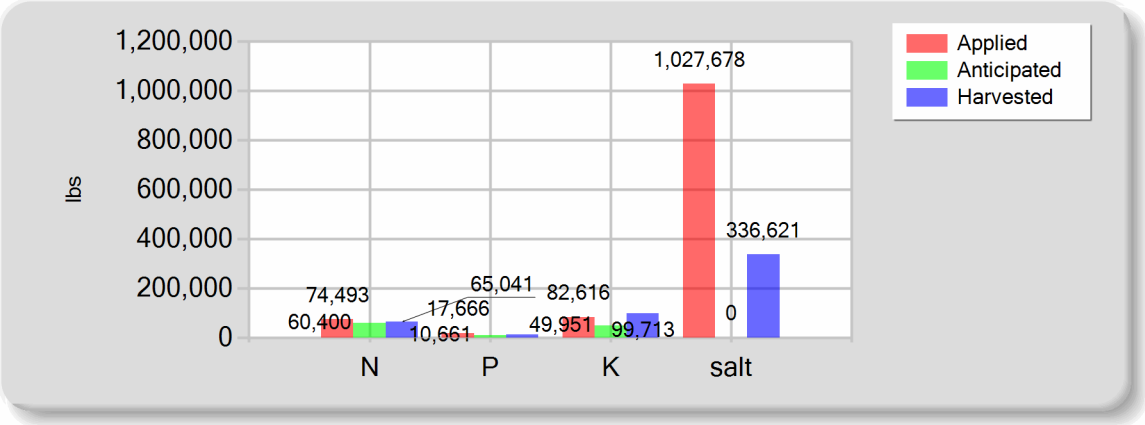
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

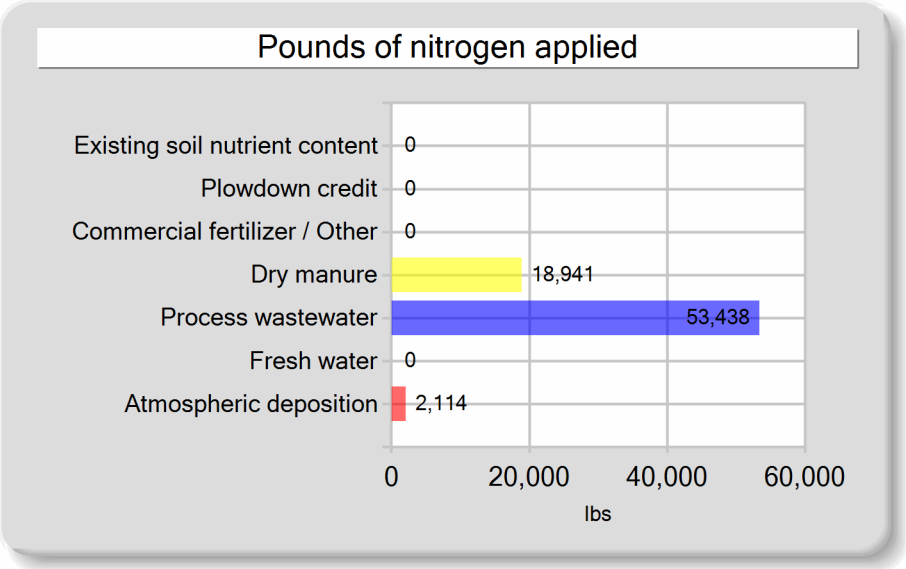
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

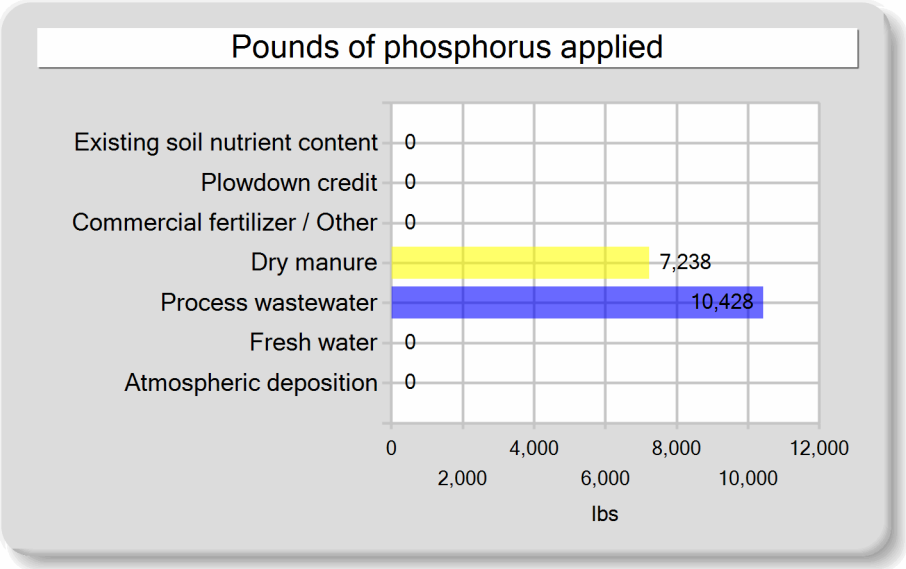
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	18,941.44	7,238.34	4,532.42	257,062.40
Process wastewater	53,437.82	10,427.56	78,083.81	731,655.01
Fresh water	0.00	0.00	0.00	38,960.77
Atmospheric deposition	2,114.00	0.00	0.00	0.00
Total nutrients applied	74,493.26	17,665.89	82,616.23	1,027,678.18
Anticipated crop nutrient removal	60,400.00	10,660.60	49,950.80	0.00
Actual crop nutrient removal	65,041.19	12,152.47	99,713.26	336,620.83
Nutrient balance	9,452.07	5,513.43	-17,097.03	691,057.35
Applied to removed ratio	1.15	1.45	0.83	3.05

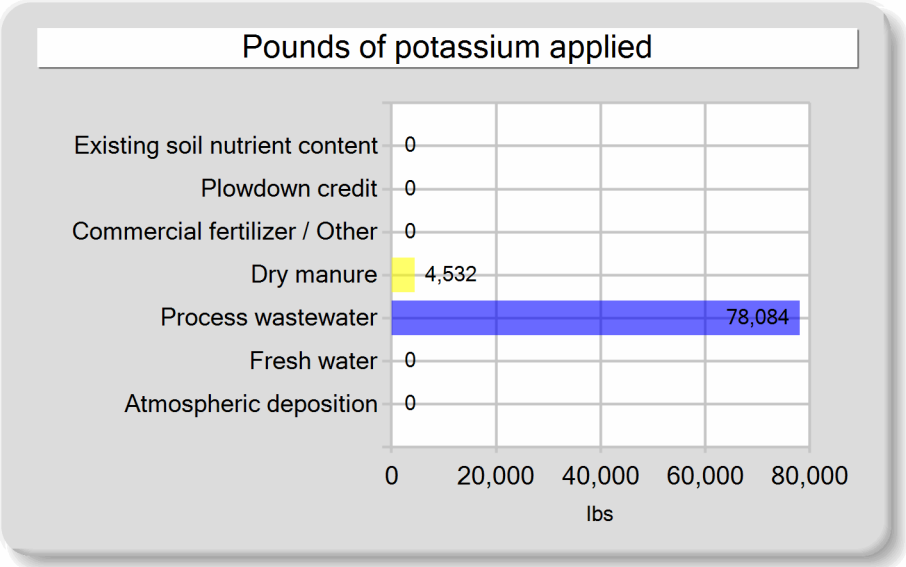
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

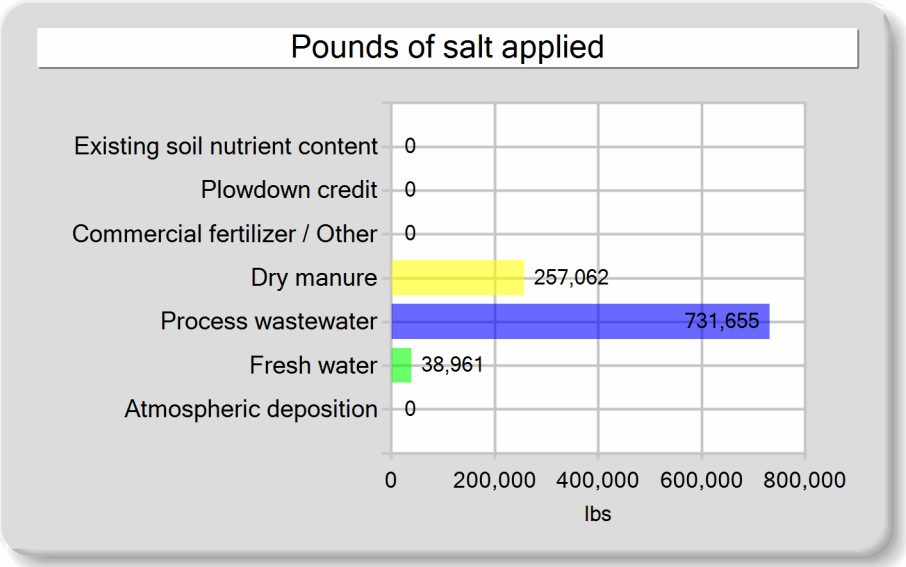


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Foy #2, Lentz and Sammon fields were planted in almonds in 2021. These fields are still not producing.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Wendel Trinkler Jr. Wendel Trinkler Jr. (May 20, 2024 09:49 PDT)	Wendel Trinkler Jr. Wendel Trinkler Jr. (May 20, 2024 09:49 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Wendel Trinkler, Jr.	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 20, 2024	May 20, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Wendel Trinkler, Jr.
P.O. Box 10
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/20/23 12:25 PM
SUBMITTED BY: Dauna
ANALYZED DATE: 10/25/23

Manure On-Site (M3)
SOURCE: Manure

Sample ID	Lab ID	Analyte	Method Ref	Result	Units	Lbs/Ton as Rec'd
Sinclear Sep. Solids	T3293021A	Moisture (H2O)	SM2540B	53.8	%	N/A
		Total Nitrogen (TN)	Combustion	1.64	%	15.2
		Total Phosphorus (TP)	SM 4500P E	0.375	%	3.47
		Total Potassium (TK)	EPA 200.7	0.597	%	5.52

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Total Nitrogen (TN)				
Total Phosphorus (TP)				
Total Potassium (TK)				

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Wendel Trinkler, Jr.
P.O. Box 10
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/20/23 12:25 PM
SUBMITTED BY: Dauna
ANALYZED DATE: 10/23/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Sammons	T3293021B	Electrical Conductivity (EC)	EPA 120.1 and	1.06	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	33.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	678	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.4	96.4 (T3293019G)	97.6 (T3293019G)	1.23
Electrical Conductivity (EC)	94.3			1.20
Nitrate Nitrogen (NO3-N)	104			3.37

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Compliance Analysis Report

Customer
Wendel Trinkler, Jr.
P.O. Box 10
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/20/23 12:25 PM
SUBMITTED BY: Dauna
ANALYZED DATE: 10/23/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Yellow House	T3293021C	Electrical Conductivity (EC)	EPA 120.1 and	1.40	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	50.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	895	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.4	96.4 (T3293019G)	97.6 (T3293019G)	1.23
Electrical Conductivity (EC)	94.3			1.20
Nitrate Nitrogen (NO3-N)	104			3.37

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Compliance Analysis Report

Customer
Wendel Trinkler, Jr.
P.O. Box 10
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/20/23 12:25 PM
SUBMITTED BY: Dauna
ANALYZED DATE: 10/23/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn West	T3293021D	Electrical Conductivity (EC)	EPA 120.1 and	1.40	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	46.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	893	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.4	96.4 (T3293019G)	97.6 (T3293019G)	1.23
Electrical Conductivity (EC)	94.3			1.20
Nitrate Nitrogen (NO3-N)	104			3.37

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Wendel Trinkler, Jr.
P.O. Box 10
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/20/23 12:25 PM
SUBMITTED BY: Dauna
ANALYZED DATE: 10/23/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn East	T3293021E	Electrical Conductivity (EC)	EPA 120.1 and	1.32	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	47.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	842	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.4	96.4 (T3293019G)	97.6 (T3293019G)	1.23
Electrical Conductivity (EC)	94.3			1.20
Nitrate Nitrogen (NO3-N)	104			3.37

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Wendel Tanker Jr.
Mailing address: PO BOX 10
(not facility address) Ceres CA 95307
Phone#: 209-537-9883

Authorized Copy Released To: F&R ag
Sampled By: Dawna Rebier
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9/21/23		Sinclear Sep Solids	manure					X	X	
2	10/17/23		Sammons	well	11.2°		X				
3	|		yellow house	|	12.4°		|			
4	|		milkbarn west	|	13.0°		|			
5	|		milk barn east	|	12.0°		|			
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)*
1st	[illegible]	Wendel Tanker			Logged in By: LABORATORY NUMBER

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Wendel Trinkler, Jr.
P.O. Box 10
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:00 AM
SUBMITTED BY: Jon
ANALYZED DATE: 12/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW #1	T3362014A	Electrical Conductivity (EC)	EPA 120.1 and	1.56	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	59.7	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	860	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	60.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	996	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	90.2	97.6 (T3362094B)	99.4 (T3362094B)			1.82
Electrical Conductivity (EC)	96.7			0.649 (T3362010E)	0.649 (Dup)	0.015
Nitrate Nitrogen (NO3-N)	104	105 (T3361023A)	103 (T3361023A)			1.64
Total Dissolved Solids (TDS)	104			560 (T3362017B)	595 (Dup)	6.06
Total Nitrogen (TN)	102			0.280 (T3353003G)	0.280 (null)	0.000

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Wendel Trinkler, Jr.
P.O. Box 10
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:00 AM
SUBMITTED BY: Jon
ANALYZED DATE: 12/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 6.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW #2	T3362014B	Electrical Conductivity (EC)	EPA 120.1 and	1.35	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	47.2	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	750	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	50.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	861	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	90.2	97.6 (T3362094B)	99.4 (T3362094B)			1.82
Electrical Conductivity (EC)	96.7			0.649 (T3362010E)	0.649 (Dup)	0.015
Nitrate Nitrogen (NO3-N)	104	105 (T3361023A)	103 (T3361023A)			1.64
Total Dissolved Solids (TDS)	104			560 (T3362017B)	595 (Dup)	6.06
Total Nitrogen (TN)	97.4			2.80 (T3362014B)	2.80 (null)	0.000

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Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Wendel Trinkler, Jr.
P.O. Box 10
Ceres, CA 95307

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:00 AM
SUBMITTED BY: Jon
ANALYZED DATE: 12/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 6.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW #3	T3362014C	Electrical Conductivity (EC)	EPA 120.1 and	1.37	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	46.0	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	765	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	46.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	879	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	90.2	97.6 (T3362094B)	99.4 (T3362094B)			1.82
Electrical Conductivity (EC)	96.7			0.649 (T3362010E)	0.649 (Dup)	0.015
Nitrate Nitrogen (NO3-N)	101	92.2 (T3362014C)	91.3 (T3362014C)			1.00
Total Dissolved Solids (TDS)	104			560 (T3362017B)	595 (Dup)	6.06
Total Nitrogen (TN)	97.4			2.80 (T3362014B)	2.80 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Wendel Trinkler Jr.
Mailing address: PO Box 10
(not facility address) Ceres, CA 9530
Phone#:

Authorized Copy Released To: F+B Ag.

Sampled By: Jon Rebien
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

Required Field Measurement

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/20/23	900A	IN #1	well water	4.8°C		H1				
2	↓	↓	IN #2	↓	6.8°C		H1				
3			IN #3		6.8°C		H1				
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st						
2nd		DENELE ANALYTICAL, INC.	12-28-23 11:00			T3362014

Required to be submitted to any agency? (YES / NO)

2023 Annual Report-Wendel Trinkler Jr.

Final Audit Report 2024-05-20

Created:	2024-05-20
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAyDWYOGQQyPfKVsvBBW7u1dN8lYF6FFQY

"2023 Annual Report-Wendel Trinkler Jr." History

Document created by Kristen Jones (kjones@fragservices.com)
2024-05-20 - 3:29:39 PM GMT

Document emailed to Dawna Trinkler (drebiero@trinklerfarms.com) for signature
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Email viewed by Dawna Trinkler (drebiero@trinklerfarms.com)
2024-05-20 - 4:48:27 PM GMT

Signer Dawna Trinkler (drebiero@trinklerfarms.com) entered name at signing as Wendel Trinkler Jr.
2024-05-20 - 4:49:38 PM GMT

Document e-signed by Wendel Trinkler Jr. (drebiero@trinklerfarms.com)
Signature Date: 2024-05-20 - 4:49:40 PM GMT - Time Source: server

Agreement completed.
2024-05-20 - 4:49:40 PM GMT

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